Journal of Biomedical Informatics 123 (2021) 103935

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Visualising and quantifying relevant parkinsonian gait patterns using 3D convolutional network

Luis C. Guayacán, Fabio Martínez *

Biomedical Imaging, Vision and Learning Laboratory (BivL2ab), Universidad Industrial de Santander, Bucaramanga (UIS), Colombia

ARTICLE INFO

Keywords:
Parkinson's Disease
Markerless
Gait analysis
Saliency maps
Retro-propagation

ABSTRACT

Parkinson's disease (PD) lacks a definitive diagnosis, with the observation of motion patterns being the main method of characterizing disease progression and planning patient treatments. Among PD observations, gait motion patterns, such as step length, flexed posture, and bradykinesia, support the characterization of disease progression. However, this analysis is usually performed with marker-based protocols, which affect the gait and localized segment patterns during locomotion. This work introduces a 3D convolutional gait representation for automatic PD classification that identifies the spatio-temporal patterns used for classification. This approach allows us to obtain an explainable model that classifies markerless sequences and describes the main learned spatio-temporal regions associated with abnormal patterns in a particular video. Initially, a spatio-temporal convolutional network is trained from a set of raw videos and optical flow fields. Then, a PD prediction is obtained from the motion patterns learned by the trained model. Finally, saliency maps, which highlight abnormal motion patterns, are obtained by retro-propagating the output prediction up to the input volume through two different stages: an embedded back-tracking and a pseudo-deconvolution process. From a total of 176 videos from 22 patients, the resulting salient maps highlight lower limb patterns possibly related to step length and speed. In control subjects, the saliency maps highlight the head and trunk posture. The proposed approach achieved an average accuracy score of 94.89%.

1. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder, affecting over 6.2 million people worldwide. According to the World Health Organization, this number will increase by more than 12 million by 2030 [1]. PD is characterized by the progressive loss of dopamine-producing neurons, a vital neurotransmitter in the execution of voluntary movements [2]. As a consequence, PD patients often present a series of movement disorders, such as slowness of movement (bradykinesia), postural instability, and body stiffness, among others [3].

To estimate the patient's motor stage and disease progression and measure a particular response to symptomatic treatment, disease stratification is performed using instruments that allow for coarse agreement, integrating several symptoms and movement disorder observations. Some common such scales include the Hoehn-Yahr (H&Y) scale, the Unified Parkinson's Disease Scale (UPDRS), or the Movement Disorder Society's Unified Parkinson's Disease Scale (MDS-UPDRS) [4]. Nevertheless, these scales strongly depend on the subjective analysis made by a medical specialist, resulting in a high variability in scale stratification (in [5] a weak level of agreement in a study with 50 patients and six different evaluators was reported). This, together with the lack of biomarkers for medical analysis, makes the diagnosis of PD a highly subjective process. In fact, it is estimated that the false-positive rate of PD diagnoses is approximately 35% at the initial diagnosis and 24% at the final diagnosis [6].

Classically, marker-based quantitative analysis of movement disorders has contributed to the mitigation of such subjectivity. These diagnostic protocols allow medical professionals to estimate and follow the locations of body landmarks, but at the cost of a considerable simplification of the gait complexity, following only a few points of interest. Additionally, these methods require strict protocols and their invasive nature can affect the naturalness of the patient's movements [7]. In particular, for PD patients, these approaches fail to recognize subtle and localized movement disorders that are strongly correlated with disease. Also, predominant parkinsonian patterns that are related to rigidity or are predominant at the initial gait phase, such as freezing gait, are not considered by classical approaches.

* Corresponding author.
E-mail address: famarcar@uis.edu.co (F. Martínez).

https://doi.org/10.1016/j.jbi.2021.103935
Received 26 March 2021; Received in revised form 5 October 2021; Accepted 10 October 2021
Available online 24 October 2021
1532-0464/© 2021 Elsevier Inc. This article is made available under the Elsevier license (http://www.elsevier.com/open-access/userlicense/1.0/).

More specialized tools to properly quantify global and localized parkinsonian patterns, as well as to discover new dynamic relationships, are required to support the diagnosis and progression monitoring of the disease. Recently, markerless methods have been proposed to capture, describe, and quantify parkinsonian gait patterns within a specific domain. For instance, common approaches based on silhouettes and RGB-D cameras (depth sensors) approximate dynamic poses from frame joint estimation, allowing the computation of typical measures such as step length, cadence, or global velocity [8,9]. Likewise, some approaches have proposed the use of video feature extraction to compute indexes or statistical information that describe PD gait [10]. These approaches build apparent motion representations from videos, which could be useful to predict and quantify abnormal patterns related to the disease. In addition, convolutional deep learning approaches have learned end-to-end features to obtain coarse predictions of different gait-related diseases [11,12].

This work introduces a new parkinsonian video representation that exploits spatio-temporal relationships during locomotion recordings to enhance the main motion and structural patterns. This representation can explain the disease from an automatic classification perspective. [1] This approach takes advantage of a deep learning net that uses volumetric and hierarchical features from 3D convolutional layers, allowing the natural processing of video sequences and capturing postural and motion patterns related to the disease. The proposed approach starts by learning a disease video representation using a supervised classification scheme. Then, a set of representative saliency maps is obtained through a retro-propagation process of the final CNN layer to the input video space. These maps represent an explainable representation that identifies the main spatio-temporal regions associated with abnormal patterns. This representation can potentially support the diagnosis of PD and subsequently be used as an indicator for physicians. Additionally, the analysis of such gait saliency regions could lead to a deeper understanding of the disease by revealing hidden patterns associated with this pathology. Interestingly, during the evaluation of the proposed approach, the saliency maps associated with PD patients enhanced spatial regions related to feet during the single support phase. The volumetric regions bounding feet in the video stand out as principal regions that contribute to the prediction of Parkinson's class. This could be related to some typical parkinsonian symptoms (see *Evaluation and Results* section).

2. Related works

Gait analysis is generally carried out using marker-based approaches, following protocols such as the 3D Mocap system, Vicon, or IR cameras, among others [14]. These protocols use passive markers (reflective landmarks), active markers (LEDs), or electronic devices (wearable sensors) placed on the patient's body to capture movement patterns during locomotion [15]. These methods are invasive, alter natural gestures, and strongly simplify locomotion dynamics to a reduced set of trajectories recorded during the gait cycle. Regarding Parkinson's characterization, these approaches can also lose sensitivity to capture, characterize, and model movements such as tremors and dystonia (simultaneous contraction of opposing muscles). Hence, several markerless methods have been proposed to complement classical gait analysis, but also to cover new and specific characteristics such as abnormal patterns related to Parkinson's disease, captured from video sequences. From the design of new descriptors, it is possible to model and understand specific dynamic gait patterns. The next subsections summarize the main markerless approaches, which could potentially be used in PD modeling.

2.1. Joint estimation approaches

Classical gait analysis is based on joint estimation during locomotion, which allows the approximation of dynamic pose configurations and measurement of particular changes during specific gait phases. Many markerless approaches have attempted to create such systems by modeling the skeleton and joints as primitives to recover biomechanical features. For instance, Khan et. al. [16] squeletonized binary silhouettes to obtain features such as the posture lean angle and stride variability, and then computed the cosine similarity with respect to an ideal gait pattern. Likewise, Das et. al. [17] generated skeletons to extract gait features, which were thereafter used to automatically classify gait diseases. For classification, the combination of the hidden Markov model and support vector machine showed promising results. In addition, some studies have taken advantage of RGB-D sensors (Kinect), adding depth information to estimate joints more accurately than the Vicon system [18,19]. With respect to this commercial system, the RGB-D approximation can follow relevant clinical movements with the same spatial accuracy except for some limitations for small movements. These methods allow the capture of joint movement without affecting the patient's natural gait. However, they are limited to emulating traditional marker-based measures, simplify gait characterization and losing relevant spatial and kinematic information.

2.2. Feature extraction approaches

Alternatively, some markerless approaches are based on modeling common gait variables and uncommon video features to build new representations that differentiate among a set of diseases, or diagnostic levels. For instance, Chien-Wen et al. encoded the mean and covariance of silhouettes to build principal component descriptors, which were thereafter combined with linear discriminant analysis to classify different gait pathologies [20]. Likewise, Shaw et al. proposed a study to inspect the gait features of 16 PD subjects and 16 normal controls using silhouettes and a hidden Markov model for disease detection [21]. However, classification using different machine learning techniques tends to generate over-fitting or under-fitting. Selecting an optimized feature is a difficult task. Thus, Bhaskar et al. proposed a system with optimal features using an analysis of variance combined with incremental feature selection. These selected features were then fed as inputs to different machine-learning methods. These individual classifiers were combined to produce a classifier fusion model [22]. These feature-based approaches have achieved significant results in parkinsonian pattern recognition; however, the results obtained with these methods do not identify a clear correlation with the symptoms and do not contribute much to the understanding of the disease.

2.3. CNN-based approaches

Recently, deep learning approaches have outperformed machine learning and other computer strategies to learn and model complex patterns that represent high-level instances, such as PD. More specifically, convolutional neural networks (CNNs) allow end-to-end modeling, enabling the discovery of visual features and the classification of samples. The CNNs build features at different levels of complexity and achieve invariant representation of actions.

Action recognition in video sequences attempts to associate video patterns with particular instances of human execution. This area could potentially be projected on the automatic analysis and support of disease classification and prediction. For instance, Simonyan et al. [23] proposed a CNN architecture that codifies raw sequences (RGB) and optical flow streams in two convolutional branches. However, that work was shown to be poorly descriptive temporally, because stacking 5 to 10

[1] A very preliminary version of this work appeared in [13]. In this extended version: 1) it was included a considerable extension of the experimental evaluation, 2) it was performed an exhaustive analysis and description of the proposed strategy, and 3) it was considered dense optical flow fields as input to reinforce kinematic gait patterns.

optical flow images did not significantly affect the results.

Additionally, recurrent neural architectures [24] have been implemented to model non-linear patterns that can be associated with complex motion sequences. These approaches could be used for discovering non-linear relationships between different and distant gait phases. Nonetheless, these approaches codify such temporal relationships from encoded vectors that lose the spatial representation of videos, which could be a determinant in examining postural configurations of the patient.

CNNs have also been implemented to identify patterns related to PD using other sensors. For instance, new learning representations have been obtained by using ECG signals [25], audio and vocal features [26], and even by modeling contrast changes in magnetic resonance sequences [27]. These models provide quantitative representations that allow a better understanding of the disease. However, these strategies depend on sophisticated protocols, and many of these strategies are invasive.

3. Proposed approach

This work introduces a PD gait representation that exploits spatio-temporal relationships during locomotion to highlight the main motion and structural patterns. This representation constitutes a potential new imaging biomarker for the disease. The main contribution of this work is a set of explainable attention videos that represent the main spatio-temporal regions associated with PD. First, a robust deep representation is achieved using a deep neural network that recovers volumetric video features from the learning of a set of 3D kernels. This convolutional network allows us to obtain a robust gait motion representation by exploiting long-term temporal relationships and, therefore, effectively captures descriptive parkinsonian gestures during walking. This architecture was evaluated from two different inputs, *i.e.*, raw values of video pixels and optical flow vector fields that describe gait in terms of dense instantaneous velocity maps. Under a supervised scheme, the convolutional net can predict and classify particular videos as PD or control gait. Once the network was trained, retro-propagation was proposed to obtain volumetric maps that explain a particular decision taken by the network to classify a video. For parkinsonian videos, these maps represent the regions where the network focuses on locating abnormal patterns. The analysis of such salient gait regions could lead to a deeper understanding of the disease and become a diagnostic support tool. The pipeline of the proposed strategy is shown in Fig. 1.

3.1. Gait volumetric sequence learning

Predominant locomotion patterns and main gait phases are described along cycles, which last an average of 1.07 seconds for healthy people [28]. In PD patients altered locomotion patterns can change the gait periodicity, and the main temporal gait markers may be described in longer time periods. For video analysis, recent approaches have attempted to capture the spatio-temporal structure of complex movements and learn action representations using convolutional neural networks. However, such representations are typically learned at the frame level, or in short intervals of time spanning a few frames, thereby failing to model complex locomotion patterns or recover main gait stages.

In this work, a 3D-convolution strategy is introduced to highlight abnormal parkinsonian gait gestures in video sequences. We implemented a long-term video representation of the spatio-temporal volumes [29]. This architecture considers spatial and long-term temporal convolutions to locally learn spatio-temporal features related to PD, allowing natural processing of video sequences and capturing relevant postural and motion patterns. Under the assumption that most relevant information is in time rather than in space (for instance, bradykineasia movements are spanned in tens of frames), the architecture at each layer increases the temporal length of representations while decreasing spatial resolution with respect to previous 3D-CNN approaches. Additionally, the simplicity of this 3D-CNN scheme keeps activations traceable back through the network. The 3D-CNN implemented herein can be understood in two different stages. First, a series of convolutional layers, each of which is followed by a process of nonlinearization (pooling), that learns gait features at different levels of complexity. The second stage describes a series of dense layers (fully connected) from which an embedded representation of the learned high-level convolutional features is obtained. These stages are described in the following subsections.

3D convolutional representation: In the first stage of the network, each layer L consists of a 3D convolutional layer followed by a max-pooling layer. This stage allows a robust representation of spatio-temporal patterns, learning salient gait characteristics at different levels, that are associated with specific entities. Let $\mathbf{I}(\mathbf{x})_\mathbf{t}$ be an image in the video sequence at time t, where $\mathbf{x} = (x, y)^T$ are spatial pixels. For each layer L, linear transformations are computed progressively, followed by contractive non-linearities, projecting the information onto a set of q learned filters, expressed as $\Psi_L = \{\psi_i\}_{i=1}^{q}$. Thus, the image $\mathbf{I}(\mathbf{x})_\mathbf{t}$ is initially filtered in the first layer by the set Ψ_1, obtaining a representation: $\Phi_1(\mathbf{x}) = \sum_{g=1\ldots q} \mathbf{I}(\mathbf{x})_\mathbf{t} * \psi_g$, where ψ_g is each independent learned

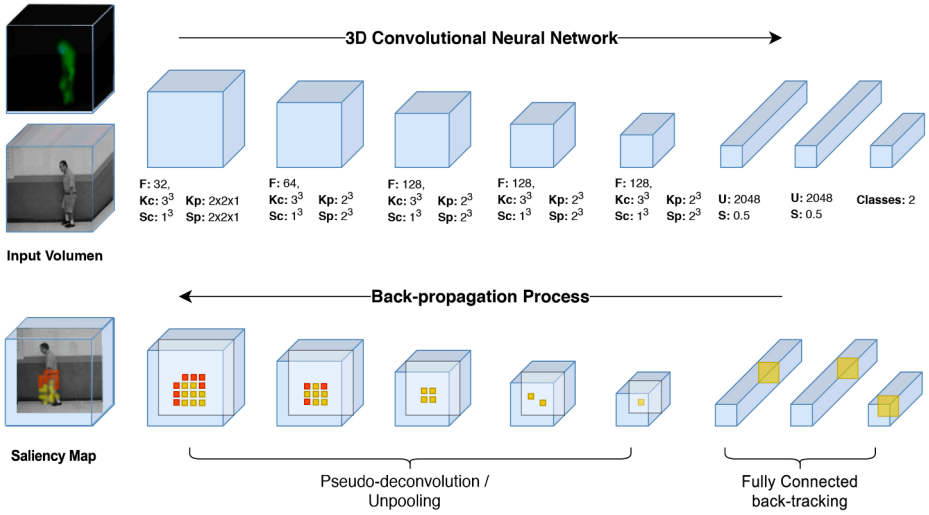


Fig. 1. Pipeline of the proposed approach. Top: a 3D convolution scheme implemented to capture spatio-temporal motion patterns related to PD. This architecture is capable of automatically classifying PD from control subjects. Bottom: a retro-propagation scheme allows us to obtain salient disease regions.

convolution filter. The resulting representation $\Phi_1(\mathbf{x})$ is then successively and progressively convoluted with the filters of the following $N-1$ layers $\{\Psi_i\}_{i=2}^{N}$. The convolutional representation is defined by the representation in each layer: $\{\Phi_i(\mathbf{x})\}_{i=1}^{N}$.

The responses in the first layers of this network generate low-level and generalized features such as borders and textures. These low-level features are combined to build more complex representations in the subsequent convolutional layers. Finally, the features found in the last convolutional layer correspond to high-level features that can represent the semantics of the data. The network proposed in this work is composed of five 3D convolutional layers, as depicted in Fig. 1.

Embedded representation: The learned hierarchical and convolutional representation is embedded in a fully connected network that learns non-linear combinations of these features and obtains a low-dimensional continuous vector representation. In these embedded vectors, the neurons have full connections to all activations of the previous layer, enabling the classification of PD. Formally, these layers can be described as follows:

$$X_L = X_{L-1} \odot W_{L-1} + B_{L-1}$$

where X is the input of the layer, W is the weights, and B is the biases. In this study, the network consisted of three dense layers. The last layer corresponds to the softmax, where the number of units is equivalent to the number of classes. This type of representations can characterize parkinsonian patterns recorded in video sequences, which can lead to the analysis of visual relationships related to the disease in an embedded compact space.

3.2. Parkinsonian patterns from volumetric saliency maps

A main contribution of this work was the analysis of spatio-temporal parkinsonian patterns obtained from the convolutional representation, understood as the main volumetric regions that contribute to a particular disease classification. For this purpose, the first scheme proposed herein is used to obtain attention saliency maps by back-tracking activations from embedded vector representations to the first volumetric layer. To do so, the activated neuron with the highest contribution to the value of the subsequent layer is analyzed using an inverse-prediction scheme. Because the proposed approach has both embedded and 3D convolutional layers, two different back-tracking schemes were conducted:

Embedded back-tracking: In this first stage, the top embedding vectors are backtracked by selecting the major index in layer L that identifies the most relevant vector on layer $L-1$. Formally, this relation can be expressed as:

$$E_{L-1} = \underset{j}{\text{argmax}} \sim \left(X_{L-1} \odot W_{L-1}^{j,E_L} \right)$$

where X is the input of layer $L-1$ with shape $1 \times n$, and W is the weights of layer $L-1$ with shape $n \times m$ (n and m represent the input and output sizes of layer $L-1$, respectively). On W, j and E_L are row and column positions, respectively. For instance, the most significant probability index at the softmax layer (E_L) points out to the maximum index on the corresponding vector of the previous layer (E_{L-1}). Finally, the back-tracked index from the flattened information is coded into volumetric indexes and propagated along channels (unflattening).

Pseudo-deconvolution/Unpooling: In this process the most relevant indexes recovered by the embedding backtracking process are projected to volumetric layers and traced back to the input video. This allows us to obtain a 3D representation of the relevant spatio-temporal information learned by the neural network. In general, unpooling is performed for each layer, followed by a pseudo-deconvolutional process. For the unpooling process, it is sufficient to read the location of the maximum pooling values, previously stored during network training. The set of unpooling indexes in layer L is denoted as $U_L = \{(t_i, h_i, w_i)\}_{i=1}^{n}$. Each unpooling index represents a region in the activation of the previous layer.

For the pseudo-deconvolution process, this region is element-wise multiplied with the set of kernels, and the maximum argument outputs are returned as the most relevant 3D positions on the previous layer. This process is then performed back to the input image, revealing the most important region that contributes to the classification label. This can be expressed as follows:

$$D_L = \bigcup_{u \in U_L} \bigcup_{\psi \in \Psi_L} \text{argmax} \left(X_L^u \odot \psi \right)$$

where Ψ_L is the set of convolutional kernels of layer L and X^u is the $3 \times 3 \times 3$ segment of tensor X centered at position u. Thereafter, this process is iteratively repeated to estimate additional points with less relevance but with influence on the classification process. These points are joined together to form the saliency maps that represent the main volumetric regions related to the disease, as depicted in Fig. 2.

3.3. Data

The dataset consists of videos captured with a conventional camera (Nikon D3200) at 60fps, under semi-controlled conditions from 22 participants, with 11 control subjects and 11 patients diagnosed with PD. The dataset consists of 12 men and 10 women distributed as follows: two women and nine men in the PD group, and eight women and three men in the control group. This dataset is age-matched, *i.e.*, the mean age is 72.2 ± 4.0 for control subjects and 72.6 ± 7.7 for PD patients. The PD patients were in stages of the disease between 2.5 and 3.0 according to the Hoehn and Yahr scale (estimated by a physical therapist). Each subject in the study was recorded eight times while performing a natural markerless walk, four times to the right and four times to the left, for a total of 176 video sequences. The entire set of videos was recorded indoors, with a static camera, a green background, and patients wearing different clothes. For the experiments, the videos were down-sampled to a length of 60 frames, cropped to 500×500 pixels, and resized to 256×256 pixels. All video sequences were recorded at the FAMPAS foundation *(Fundación del Adulto Mayor y Parkinson Santander)* and at the nursing home *Asilo San Rafael*. This study was approved by the Ethics Committee of the Universidad Industrial de Santander, and written informed consent was provided by all subjects.

Optical flow: The analysis of motion patterns as a velocity dense field has been a common method in multiple approaches to action recognition. In this work, we implement *Flownet 2.0*, an optical flow approach based on CNNs that has exhibited a significant improvement in

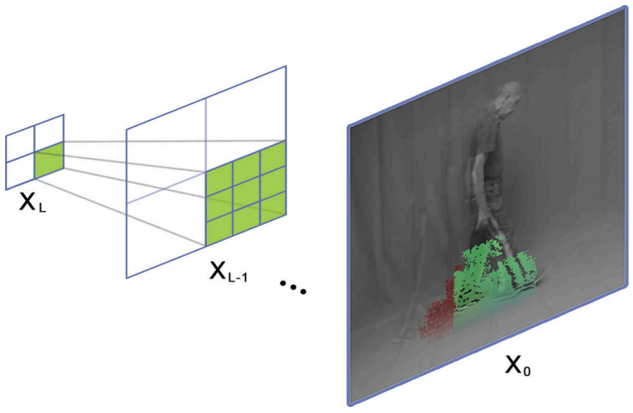


Fig. 2. Pseudo-deconvolution during retro-propagation. The image illustrates how, through the pseudo-deconvolution process, each point obtained during the embedded back-tracking stage in layer X_L represents a region on the patient's body in the original sequence X_0.

quality compared with traditional methods [30]. This optical flow is an improvement of a previous CNN approach called *Flownet* where the authors simply stacked two sequentially adjacent input images together and fed them through the network. They also presented another variation where they first produced representations of the two images separately, before combining them together in a *correlation layer*, thereby learning the higher representation together. *Flownet 2.0* introduced several significant improvements, using a stacked architecture and introducing a specialized sub-network dedicated to capture and code small motions. Specifically, *Flownet 2.0* implements a stacked architecture where the first network obtains images I_1 and I_2 as input. Subsequent networks obtain I_1, I_2, and the previously estimated flow $w_i = (u_i, v_i)^T$ where i denotes the index of the network in the stack. Additionally, they warp the second image $I_2(x, y)$ via $\widetilde{I}_{2,i}(x, y) = I_2(x + u_i, y + v_i)$ and incorporate the error $e_i = \| \widetilde{I}_{2,i} - I_1 \|$. As a result, Flownet 2.0 presented a large improvement in quality as well as speed (see an example in Fig. 3). We hypothesize that such an approach can be effective in recovering motion patterns associated with PD, capturing coarse motion in legs while also paying attention to small variations probably associated with tremors.

3.4. Experimental setup

The proposed architecture follows the layer configuration shown in Fig. 1. In general, this strategy has five 3D convolutional layers. The number of kernels varied between 32 and 128. At the end, two dense vectors enable an embedding representation of the videos. During the learning process, stochastic gradient descent was used to minimize the loss function. The best performance was obtained with 10 and 20 epochs for RGB and optical flow sequences, respectively, with a batch size of two. All experiments were initialized with a momentum of 0.9, dropout of 0.5, weight decay of 5×10^{-3}, learning rate of 10^{-3}, and batch normalization. Faster convergence was achieved by using a learning rate and weight decay reduction (a factor of 10^{-1}) for the five and nine epochs.

The evaluation of the proposed approach was carried out using a leave-one-patient-out cross-validation scheme, *i.e.*, the learning algorithm was applied once for each patient, using all other patients as a training set (168 videos) and using the selected patient as the test set (eight videos). This validation scheme ensures that the proposed approach learns abnormal locomotion patterns rather than the similarities of the patient's visual appearance. Thus, a prediction was obtained for each of the 176 videos in the dataset, but run at different cross-validation folds. Once all predictions were computed in the experimental setup, the accuracy, precision, and recall were computed for all video predictions by comparison with the reference expert classification. To obtain saliency maps, the retro-propagation process was carried out three times to obtain video regions with three degrees of relevance.

4. Evaluation and results

The first evaluation of the proposed strategy was dedicated to measuring the capability of recognizing PD patients with respect to the control population. Table 1 shows the quantitative classification results of the PD and control groups from raw RGB sequences and optical flow. The RGB sequences consistently outperformed the optical flow, achieving a classification score of 94.89% compared to 84.66% for the optical flow. Interestingly, for the optical flow, the precision (0.87) was higher than the recall (0.78), in contrast to the RGB sequences with a precision and a recall of 0.91 and 1.0, respectively. However, it should be noted that in our case, a false positive would not have as many unfavorable repercussions as a false negative, whereby the diagnosis and treatment of the patient would be delayed. Thus, recall is a more meaningful measure for evaluating the proposed method.

Taking this into account, a more detailed analysis was performed by considering the F2-score. This measure weighs recall higher than precision and is expressed as $F_\beta = \left(1 + \beta^2\right).\frac{precision \cdot recall}{(\beta^2 \cdot precision) + recall}$, with $\beta = 2$, resulting in 0.70 and 0.78 for the optical flow and RGB sequences, respectively.

Additionally, the ROC curve was quantified, as illustrated in Fig. 4 to show a good trade-off between sensitivity and specificity for the proposed model. The proposed approach achieves an area under the curve (AUC) of 0.95 for RGB sequences and 0.91 for optical flow. Fig. 5 illustrates the corresponding confusion matrices. Considering that there are eight videos of each patient, it is important to note that the model for the RGB images completely misclassified only one patient (outlined in red in Fig. 6). This patient corresponds to a female patient of the control class who shows decreasing functions related to gait, such as speed, cadence, stride, and foot angle, except for support base, which seems to be higher. These gait patterns, which were "correctly detected" by the model as abnormal, could be related to the patient's overweight.

The second evaluation is related to the ability to detect salient regions associated with disease patterns. In this case, we follow a principle of model explainability, which attempts to retro-propagate a particular decision over previous layers to detect the main regions associated with a given score. Saliency maps are obtained by retro-propagating the output prediction up to the input volume. Different saliency maps can be obtained by varying the maximum to be taken in the penultimate layer of the CNN. In Fig. 6, red represents the first maximum, green represents the second maximum, and blue represents the third maximum value considered in this layer. In this sense, red, green, and blue colors represent the degree of importance of each region (decreasingly).

The method is able to locate and highlight spatio-temporal patterns on the patient's body. Interestingly, in most of the parkinsonian gait sequences, the proposed method highlights the lower body during the

Table 1
Summary of the quantitative classification result for raw RGB and Optical flow (OF) sequences.

	Accuracy (%)	Precision	Recall	F2-score
RGB	**94.89**	0.91	1.00	0.78
OF	84.66	0.87	0.78	0.70

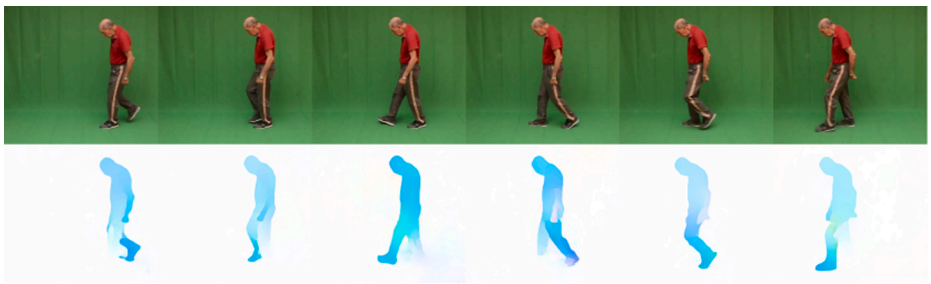


Fig. 3. First row: Image from the original video sequence of a patient with PD. Second row: Optical flow computed using Flownet2.0.

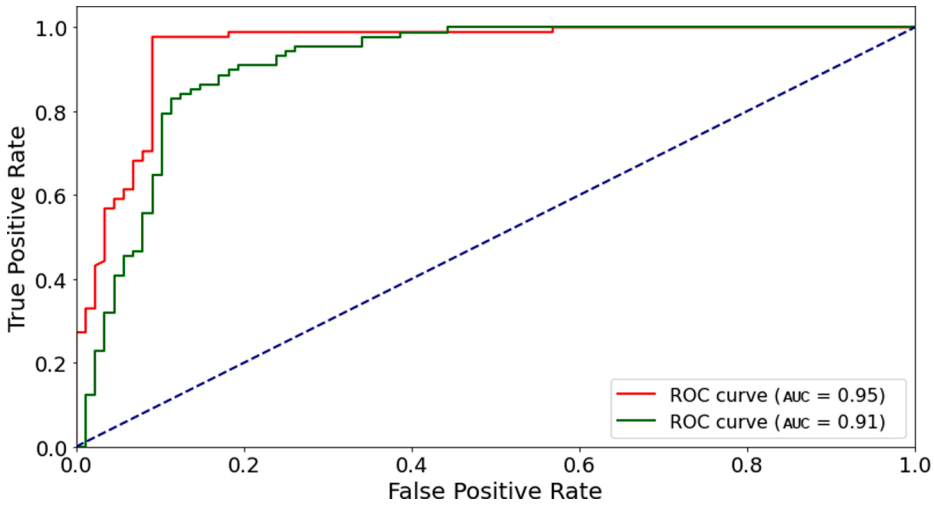


Fig. 4. ROC curve for RGB and optical flow sequences. Raw videos outperform optical flow with an AUC value of 0.95 vs. 0.91.

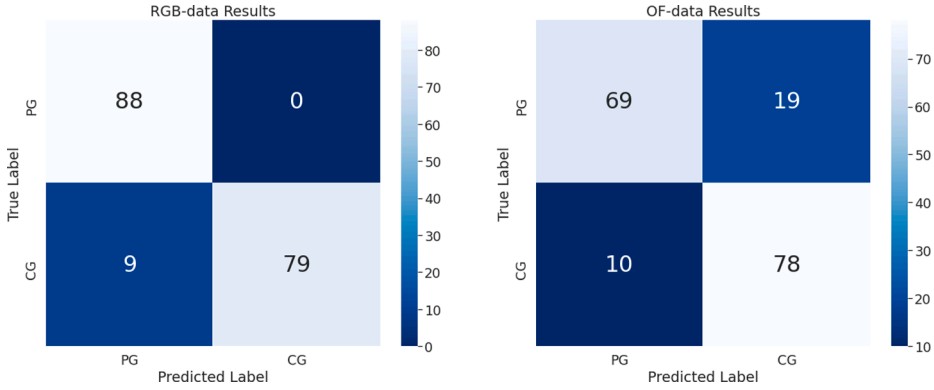


Fig. 5. Confusion matrices for RGB (left) and optical flow data (right).

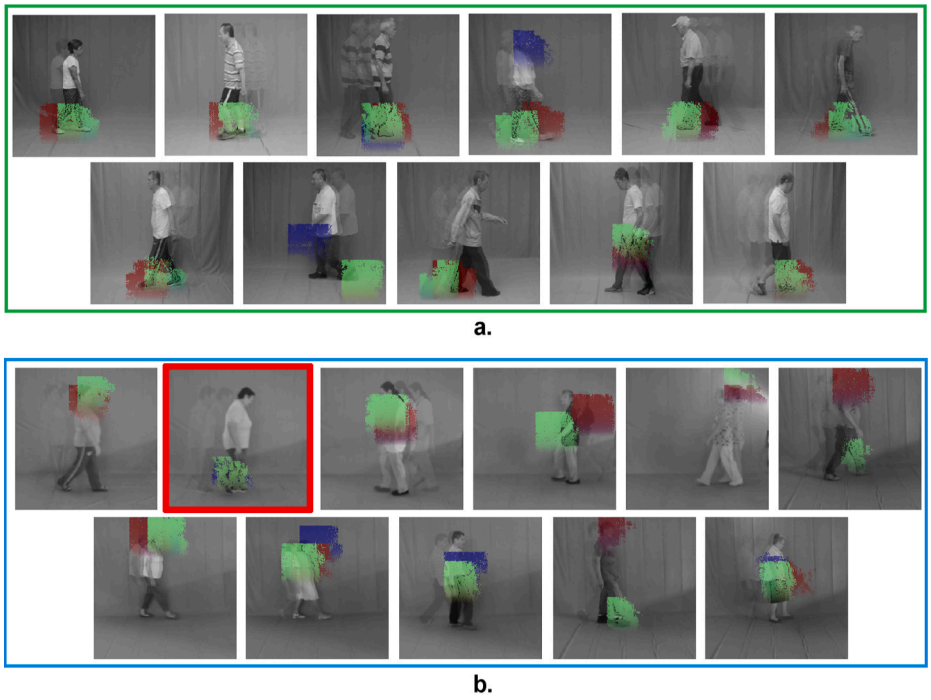


Fig. 6. Salient motion maps for PD patients (a) and control subjects (b).

single support phase, whereas in the control gaits, the method highlights the upper part of the body. This suggests that the model finds abnormal spatio-temporal features in the legs of patients with PD (possibly related to stride length, cadence, or foot clearance) and identifies healthy features related to the motion of the head and the straight posture of the control subjects. These patterns constitute relevant information that can potentially support physician observations.

Several state-of-the-art methods for the computation of saliency maps from CNNs have been proposed where the heat map is a weighted combination of feature maps. For example, GradCam [31] computes the gradient of the score for a given class with respect to the feature maps of a convolutional layer. These back-propagated gradients were global-average-pooled to obtain the importance weights. However, these methods require that there is no significant dimensionality reduction in the convolutional layers. Additionally, these methods process one image at a time and, to the best of our knowledge, there is no method for the direct calculation of saliency maps in spatio-temporal volumes. Finally, as depicted in Fig. 7, the association with disease motion patterns in our maps is convincing. In contrast, the maps that result from GradCam mainly focus on background and normalize the region probabilities within the scene. As a result, these baselines lack the capability to visually explain a particular prediction. The maps achieved for the proposed approach remain coherent in time and can be useful visual tools for medical diagnosis support.

4.1. Confounding effects

The data set used in this work consisted of 12 men and 10 women distributed as follows: two women and nine men in the PD group and eight women and three men in the control group. The study was limited to participants available at the clinical center, and the gender data imbalance may be associated with a higher incidence of disease in men than in women [14]. Therefore, a confounding analysis was performed to determine the influence of this dataset mismatch on the results obtained. Gender and age were evaluated as possible confounding variables by computing crude and adjusted odds ratios using a logistic regression process. In this model, the predictor variables x_i are linearly related to the log-odds of the outcome event y. For three predictors, this is formally expressed as follows:

$$ln\left(\frac{P(y=1)}{1-P(y=1)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

where y is the predicted probability of PD, x_1 is the binary true label, x_2 is the gender and x_3 is the age of each participant. For the crude odds ratio (cOR), the regression coefficient β_1 is calculated without considering the effect of the other coefficients, that is, $\beta_2 = \beta_3 = 0$. For the adjusted odds ratio (aOR), β_1 is computed along with the other coefficients. The odds-ratio estimation is then defined as an exponential function e^{β_1} [32].

The obtained cOR and aOR were 9.38 and 8.13, respectively, ($p < 0.05$). The odds ratios reflect a significantly positive association between disease diagnosis and the probabilities predicted by the proposed method. The performance of the proposed approach seems to be mainly associated with the proper modeling of spatio-temporal patterns in video. Such an association is not confounded by gender or age, indicating that these variables do not influence the results of the evaluation of the considered dataset.

5. Discussion and concluding remarks

This work introduced a novel markerless gait representation that exploits the ability of 3D CNNs to learn spatio-temporal features, allowing the visualization of key biomarkers of parkinsonian gait. This model implements long-term temporal convolutions, which makes it possible to use video sequences long enough to capture at least one complete gait cycle. This is relevant because abnormal patterns may occur at specific stages of the cycle in each patient. The trained model is able to predict and classify abnormal patterns, achieving accurate PD differentiation from control gaits. Additionally, the main contribution of this work is to obtain salient maps from a retro-propagation process through the previously trained model. This process consists of two main steps: first, the tracking of maximum activation values in the previous dense layers; and second, a pseudo-deconvolution process that finally achieves a projection on the video of the most important features for classification learned during training.

The recovered explainable attention videos show abnormal spatio-temporal patterns in the lower limbs of patients with PD during the single support phase. In this sense, the patterns recognized by the proposed model may be related to already known symptoms of PD or to unexplored hidden patterns. For instance, most PD patients in Fig. 6 exhibit a reduction in features such as stride length, cadence, or foot clearance, which corresponds to the position of the recovered saliency maps on the feet. It should be noted that this study has limitations regarding the detailed diagnosis description of each motor component for each patient. Future work may include a personalized evaluation of the motor symptoms of each patient and their corresponding saliency maps. This would allow us to determine with greater certainty whether there is a direct relationship between salient maps and the symptomatology of each patient, or whether the maps indicate unknown hidden patterns that may be valuable for the characterization and classification of PD.

For a dataset of 22 patients, the proposed approach achieved a high discrimination power for PD classification with an average classification score of 94.89% for raw RGB sequences and 84.66% for optical flow. We hypothesized that optical flow could be effective in recovering movement patterns associated with PD, capturing coarse body movements as well as the small variations probably associated with tremors. However, the results showed that there was not a significant gain with the exclusive use of optical flow patterns. This fact may be due to the coarse recovery of motion patterns or numerical errors associated with the flow field estimation. In contrast, the proposed approach effectively captures

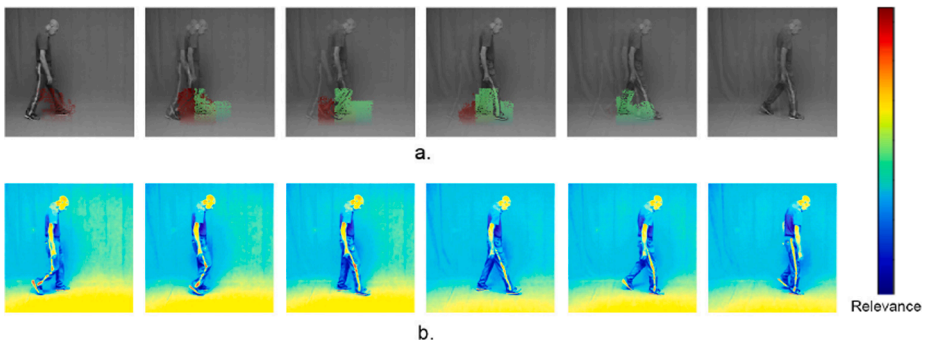


Fig. 7. Comparison between proposed method (a) and GradCam (b) saliency maps.

spatio-temporal patterns from raw videos, learning a dense and hierarchical representation of motion, which is promising for supporting PD. In addition, the raw video sequences have coded structural information from poses during locomotion that may provide extra information in classification, such as trunk inclination angle or knee flexion. On the other hand, it is possible that the implemented optical flow is not as descriptive in our particular case, so future work should evaluate other optical flow alternatives that could complement the analysis.

In clinical practice, PD is quantitatively determined from classical gait analysis, which includes the approximation of body dynamics from a set of markers. However, these marker-based methods require strict protocols that affect the naturalness of the patient's movements and are insufficient as diagnostic tools for PD. This, together with the lack of biomarkers that fully support medical analysis, makes the diagnosis of PD a highly subjective process. On the other hand, the ability to explain output predictions in a deep learning model is currently one of the most important issues in biomedical applications, to better support clinical decisions, highlighting the spatial regions that primarily contribute to classification. The proposed approach proved to be effective in classifying parkinsonian patterns with respect to control gaits and in recovering saliency maps as indicators of abnormal patterns, which could support physician observations and eventually be a decisive PD biomarker.

Although there are other methods for computing saliency maps in images, our method allows the computation of saliency maps directly in videos, taking advantage of the spatio-temporal patterns learned by the 3D convolutional network. In addition, our method appears to be more intuitive and temporally precise. In such a sense, this strategy may be introduced in diagnostic protocols to complement and support classical evaluations.

Since this work was performed using a new dataset with particular characteristics (markerless video sequences), the main limitation is the baseline comparison. The dataset considered in this study is publicly available at https://bivl2ab.uis.edu.co/datasets.

Parkinsonian patients who participated in this study were characterized in a middle or advanced stage of the disease. Future work will test this descriptor in the early stages, and a longitudinal study will be performed to characterize the evolution of the disease.

Compliance with ethical standards

This research study was conducted retrospectively using human subject data. Approval was granted by the Ethics Committees of Universidad Industrial de Santander in Bucaramanga, Colombia.

CRediT authorship contribution statement

Luis C. Guayacán: Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **Fabio Martínez:** Conceptualization, Supervision, Writing – review & editing, Project administration, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors thank the FAMPAS Foundation *(Fundación del Adulto Mayor y Parkinson Santander)* and the nursing home *Asilo San Rafael*. Additionally, grateful acknowledgments to the *Vicerrectoría de Investigación y Extensión* of the Universidad Industrial de Santander for supporting this research registered by the project: *Cuantificación de patrones locomotores para el diagnóstico y seguimiento remoto en zonas de dificil acceso*, with SIVIE code 2697.

References

[1] T. Vos, A.A. Abajobir, K.H. Abate, C. Abbafati, K.M. Abbas, F. Abd-Allah, R. S. Abdulkader, A.M. Abdulle, T.A. Abebo, S.F. Abera, et al., Global, regional, and national incidence, prevalence, and years lived with disability for 328 diseases and injuries for 195 countries, 1990–2016: a systematic analysis for the global burden of disease study 2016, The Lancet 390 (10100) (2017) 1211–1259.

[2] H. Bernheimer, W. Birkmayer, O. Hornykiewicz, K. Jellinger, F. Seitelberger, Brain dopamine and the syndromes of parkinson and huntington clinical, morphological and neurochemical correlations, J. Neurol. Sci. 20 (4) (1973) 415–455.

[3] R.B. Davis III, S. Ounpuu, D. Tyburski, J.R. Gage, A gait analysis data collection and reduction technique, Human Movement Sci. 10 (5) (1991) 575–587.

[4] C.G. Goetz, B.C. Tilley, S.R. Shaftman, G.T. Stebbins, S. Fahn, P. Martinez-Martin, W. Poewe, C. Sampaio, M.B. Stern, R. Dodel, et al., Movement disorder society-sponsored revision of the unified parkinson's disease rating scale (mds-updrs): scale presentation and clinimetric testing results, Movement Disorders: Off. J. Movement Disorder Soc. 23 (15) (2008) 2129–2170.

[5] B. Post, M.P. Merkus, R.M. de Bie, R.J. de Haan, J.D. Speelman, Unified parkinson's disease rating scale motor examination: are ratings of nurses, residents in neurology, and movement disorders specialists interchangeable? Movement Disorders: Off. J. Movement Disorder Soc. 20 (12) (2005) 1577–1584.

[6] H. Splete, Avoid diagnostic pitfalls for parkinson's disease, Internal Med. News 39 (17) (2006), 31–31.

[7] E. Ceseracciu, Z. Sawacha, C. Cobelli, Comparison of markerless and marker-based motion capture technologies through simultaneous data collection during gait: proof of concept, PloS One 9 (3) (2014) e87640.

[8] T.T. Verlekar, L.D. Soares, P.L. Correia, Automatic classification of gait impairments using a markerless 2d video-based system, Sensors 18 (9) (2018) 2743.

[9] P. Ren, J.F.B. Bayard, L. Dong, J. Chen, L. Mao, D. Ma, M.A. Sanchez, D. M. Morejon, M.L. Bringas, D. Yao, et al., Multivariate analysis of joint motion data by kinect: Application to parkinson's disease, IEEE Trans. Neural Syst. Rehabil. Eng. 28 (1) (2019) 181–190.

[10] K. Sugandhi, F.F. Wahid, G. Raju, Feature extraction methods for human gait recognition–a survey, in: International Conference on Advances in Computing and Data Sciences, Springer, 2016, pp. 377–385.

[11] R. San-Segundo, H. Navarro-Hellín, R. Torres-Sánchez, J. Hodgins, F. De la Torre, Increasing robustness in the detection of freezing of gait in parkinson's disease, Electronics 8 (2) (2019) 119.

[12] T.T. Verlekar, P. Lobat, L.D. Soares, Using transfer learning for classification of gait pathologies, in: 2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2018, pp. 2376–2381.

[13] L.C. Guayacán, E. Rangel, F. Martínez, Towards understanding spatio-temporal parkinsonian patterns from salient regions of a 3d convolutional network, in: 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2020, pp. 3688–3691.

[14] N. Kour, S. Arora, et al., Computer-vision based diagnosis of parkinson's disease via gait: A survey, IEEE Access 7 (2019) 156620–156645.

[15] D.H. Sutherland, The evolution of clinical gait analysis: Part ii kinematics, Gait & Posture 16 (2) (2002) 159–179.

[16] T. Khan, J. Westin, M. Dougherty, Motion cue analysis for parkinsonian gait recognition, Open Biomed. Eng. J. 7 (2013) 1.

[17] D. Das, Human gait classification using combined hmm & svm hybrid classifier, in: 2015 International Conference on Electronic Design, Computer Networks & Automated Verification (EDCAV), IEEE, 2015, pp. 169–174.

[18] Q. Li, Y. Wang, A. Sharf, Y. Cao, C. Tu, B. Chen, S. Yu, Classification of gait anomalies from kinect, The Visual Comput. 34 (2) (2018) 229–241.

[19] D. Tan, Y.-H. Pua, S. Balakrishnan, A. Scully, K.J. Bower, K.M. Prakash, E.-K. Tan, J.-S. Chew, E. Poh, S.-B. Tan, et al., Automated analysis of gait and modified timed up and go using the microsoft kinect in people with parkinson's disease: associations with physical outcome measures, Med. Biol. Eng. Comput. 57 (2) (2019) 369–377.

[20] C.-W. Cho, W.-H. Chao, S.-H. Lin, Y.-Y. Chen, A vision-based analysis system for gait recognition in patients with parkinson's disease, Expert Syst. Appl. 36 (3) (2009) 7033–7039.

[21] L. Shaw, Hmm based parkinson's detection by analysing symbolic postural gait image sequences, Int. J. Tech. Res. Appl 2 (4) (2014) 211–216.

[22] V.B. Semwal, J. Singha, P.K. Sharma, A. Chauhan, B. Behera, An optimized feature selection technique based on incremental feature analysis for bio-metric gait data classification, Multimedia Tools Appl. 76 (22) (2017) 24457–24475.

[23] K. Simonyan, A. Zisserman, Two-stream convolutional networks for action recognition in videos, in: Advances in Neural Information Processing Systems, 2014, pp. 568–576.

[24] J. Donahue, L. Anne Hendricks, S. Guadarrama, M. Rohrbach, S. Venugopalan, K. Saenko, T. Darrell, Long-term recurrent convolutional networks for visual recognition and description, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 2625–2634.

[25] S.L. Oh, Y. Hagiwara, U. Raghavendra, R. Yuvaraj, N. Arunkumar, M. Murugappan, U.R. Acharya, A deep learning approach for parkinson's disease diagnosis from eeg signals, Neural Comput. Appl. (2018) 1–7.

[26] H. Gunduz, Deep learning-based parkinson's disease classification using vocal feature sets, IEEE Access 7 (2019) 115540–115551.

[27] F.J. Martinez-Murcia, A. Ortiz, J.M. Górriz, J. Ramírez, F. Segovia, D. Salas-Gonzalez, D. Castillo-Barnes, I.A. Illán, A 3d convolutional neural network approach for the diagnosis of parkinson's disease, in: International Work-Conference on the Interplay Between Natural and Artificial Computation, Springer, 2017, pp. 324–333.
[28] M.P. Murray, A.B. Drought, R.C. Kory, Walking patterns of normal men, JBJS 46 (2) (1964) 335–360.
[29] G. Varol, I. Laptev, C. Schmid, Long-term temporal convolutions for action recognition, IEEE Trans. Pattern Anal. Machine Intell. 40 (6) (2017) 1510–1517.
[30] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, T. Brox, Flownet 2.0: Evolution of optical flow estimation with deep networks, in: Proceedings of the IEEE conference on Computer Vision and Pattern Recognition, 2017, pp. 2462–2470.
[31] R. Rs, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: visual explanations from deep networks via gradient-based localization, Int. J. Comput. Vis. 128 (2019).
[32] M. Szumilas, Explaining odds ratios, J. Can. Acad. Child Adolescent Psychiatry 19 (3) (2010) 227.